PS 02: Early-Warning System for Corporate Distress and Fraud ("would it have caught IL&FS?")

Generated by *python run_all.py*. Every number below is read from *outputs/polish_metrics.json* and *outputs/india_eval*.json* from the same run.

1. Summary

- **Distress model** (UCI Polish bankruptcy data, 43,405 firm-years): a monotone LightGBM on 28 size-free ratios that can be rebuilt from Indian filings, with train-median imputation. On the untouched holdout, **PR-AUC is 0.369** against a no-skill rate of 0.069. The holdout has 410 defaults in 5,910 firm-years (6.94%, about 1:13.4). ROC-AUC is 0.823, and recall at a 5% alert rate is 0.341.
- **Indian backtest.** 15 documented non-financial defaulters, 44 sector peers and a frozen universe of 893 listed companies (930 eligible borrowers at 2026-09-30). Alerts are hard-capped at min(40, 5%) per quarter.
- Headline composite recall is **1/15 [0.00-0.32] at 4 quarters** and **0/15 [0.00-0.22] at 8 quarters** before the event.
- **IL&FS was not caught.** IL&FS Transportation and IL&FS Engineering ranked 409/739 (cap 36), 136/739 (cap 36) at 4Q.
- **Honest bottom line.** Annual book ratios, available about 6 months after year end and run through a model trained on Polish SMEs, do not separate large Indian group defaults from the many small distressed listed companies that fill a 5% alert budget. Most of the warning for these cases came from governance and market events (rating downgrades, auditor exits, results delays, pledges). These are researched for the cases (Section 5.3) and built into the escalation override. They are not counted in recall, because they were not collected for non-defaulters.

2. Data

Polish (UCI 365). File k holds ratios with bankruptcy status after (6-k) years. **Horizon split:** train on 1-3year, validate on 4year (2-year horizon, about 8Q; used for selection, isotonic calibration and threshold), and keep 5year (1-year horizon, about 4Q) as the untouched holdout. This is a horizon split, not calendar time, because the files carry no dates. Duplicate rows were removed. A secondary firm-disjoint split within 5year gives PR-AUC 0.558.

split	rows	defaults	default rate
train	27703	1166	0.0421
val	9792	515	0.0526
holdout	5910	410	0.0694

Indian statements. Annual statements come from screener.in, which compiles annual reports and exchange filings (consolidated where available, otherwise standalone). There are 302,258 data points in `outputs/provenance_financials.csv.gz`. Each has a URL, an access time in UTC and an **as-of availability date = FY end + 183 days** (the Companies Act s.96 AGM deadline). 0.3% of the cells are blank at source; these are marked missing and never filled. Companies skipped as unparsable: DUMMYHEG, DUMMYINGL1, DUMMYINGL2, DUMMYINXGN, JISLDVREQS, SGL.

Default events. Each event date is the earliest default or D rating found:

company	event_date	event_type	source_url
Rolta India Ltd	2016-05-16	Missed USD bond coupon; S&P cut to D	https://www.business-standard.com/article/pti-stories/s-p-downgrades-rolta-credit-rating-to-default-116062000499_1.html
Reliance Communications Ltd	2017-05-30	CARE D and ICRA D	https://www.careratings.com/upload/CompanyFiles/PR/Reliance%20Communications%20Limited-05-30-2017.pdf
McNally Bharat Engineering Co Ltd	2018-03-13	CARE D (issuer not cooperating)	https://www.careratings.com/upload/CompanyFiles/PR/Mcnally%20Bharat%20Engineering%20Company%20Limited-12-03-2020.pdf
Videocon Industries Ltd	2018-06-06	NCLT admission CP(IB)-02/MB/2018 (SBI petition)	https://www.casemine.com/judgement/in/5c6d0eeb4a93262639683a10
Ruchi Soya Industries Ltd (now Patanjali Foods)	2017-08-31	Default date cited in creditor petition; NCLT admitted 2017-12-15	https://ibbi.gov.in/webadmin/pdf/order/2017/Dec/15th%20Dec%202017%20in%20the%20matter%20of%20Ruchi%20Soya%20Industries%20Ltd.%20CP%201371%20&%20CP%201372-I&BP-NCLT-MAH-2017_2017-12-22%2012:33:39.pdf
IL&FS Transportation Networks Ltd	2018-10-01	ICRA D	https://nsearchives.nseindia.com/corporate/jyotsnam_03102018201837_Revised_Ratings_by_ICRA_Limited_298.pdf
IL&FS Engineering & Construction Co Ltd	2018-10-05	CARE D (debt not serviced since Sep 2018)	https://www.ilfsengg.com/News/comm2exchange/SE_Intimation_Revision_of_Credit_Rating_October_6_2018.pdf
Jet Airways (India) Ltd	2019-01-02	ICRA D (payment due 2018-12-31 delayed)	https://www.icra.in/Rating/ShowRationalReportFilePdf/76249
Sintex Industries Ltd	2019-06-11	NCD principal + interest default; CARE C->D	https://sintex.in/wp-content/uploads/2025/05/Interest_non_payment_SE_SIL.pdf
Cox & Kings Ltd	2019-06-26	Commercial paper default Rs 150 cr; CARE D 2019-06-28	https://nsearchives.nseindia.com/corporate/COX_KINGS_27062019205241_BSELetterCP_452.pdf
Coffee Day Enterprises Ltd	2019-09-10	ICRA D (from BB+)	https://www.icra.in/Rating/GetRationalReportFilePdf?id=129349
Reliance Infrastructure Ltd	2020-01-23	CARE D on NCDs	https://www.careratings.com/pdf/resources/Annexure%20VI%2013-5-2020.pdf

company	event_date	event_type	source_url
Simplex Infrastructures Ltd	2020-12-10	CARE D on NCDs	https://www.careratings.com/upload/CompanyFiles/PR/202311121117_Simplex_Infrastructures_Limited.pdf
Future Enterprises Ltd	2021-01-04	NCD interest default; CARE D Jan 2021	https://www.careratings.com/upload/CompanyFiles/PR/07012021113347_Future_Enterprises_Limited.pdf
Future Retail Ltd	2021-12-31	Missed Rs 3494.56 cr OTR payment; CARE D 2022-01-03	https://www.careratings.com/upload/CompanyFiles/PR/03012022065309_Future_Retail_Limited.pdf
Reliance Capital Ltd	2019-09-20	CARE D	http://www.careratings.com/upload/CompanyFiles/PR/Reliance%20Capital%20Limited-09-20-2019.pdf

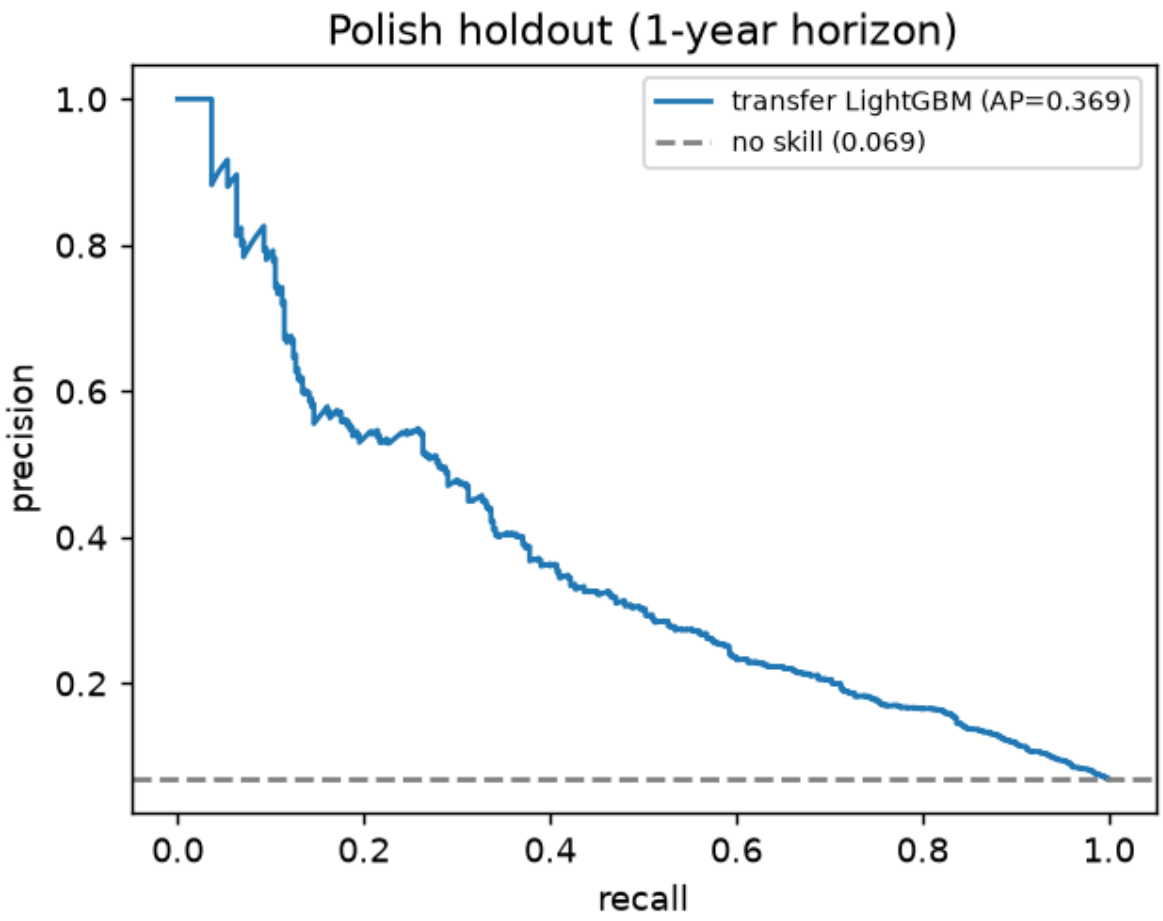
Reliance Capital (an NBFC) is a separate financial track and is not counted in recall. The IL&FS parent and IFIN, DHFL, Bhushan, Essar, Monnet and Uttam Galva have no retrievable listed-company statement pages, so they are excluded. Jaypee Infratech, Amtek and Alok (2015 events) predate peer coverage and are also excluded.

Peers and universe. 44 sector-matched peers (`peers.csv`). The universe is 598 Nifty Total Market non-financial companies plus 300 seeded-random older NSE listings, frozen on 2026-10-29. Universe size by quarter: 62-930 names. Quarters before 2015-09-30 (fewer than 300 universe names) are **not evaluable (n/e)**, and no alerts are raised in them.

3. Method and leakage controls

- **Preprocessing fit on Polish train only**: winsorise at 1/99 percentiles, then median-impute (and standardise for the logistic model). Model selection and calibration use validation only. The holdout is used once. No Indian outcome is used in any fit. Statements are used only after FY end + lag, case quarters on or after the event are dropped, and the alert cutoff is a rank, so it uses no outcomes. Automated checks: {'feature_rows_available_before_quarter': True, 'no_case_quarter_on_or_after_event': True, 'model_trained_on_polish_only': True, 'preprocessing_fit_on_polish_train_only': True, 'threshold_uses_no_outcomes': True, 'avail_lag_days': 183, 'n_quarter_rows': 37955, 'n_companies_scored': 952}.
- **Model.** LightGBM with monotone sign constraints was compared with logistic regression (balanced, C=0.5). LightGBM was selected on validation PR-AUC (0.236 vs 0.148).
- **Transfer fixes, made from covariate comparison and before the frozen run.** (a) In the Polish data, missingness itself predicts bankruptcy: holdout PR-AUC with native NaN is 0.561 (full64: 0.823). In Indian data, a missing value means a zero balance or an absent schedule, so the deployed model median-imputes. Zero finance cost now means uncapped interest cover. (b) Attr6 (retained earnings/TA) was dropped: its Polish median is 0.000, while Indian "reserves" include securities premium. Both fixes were found after a first run put debt-free IT companies at the top of the ranking. The fixes do not use the case labels, but they are disclosed.
- **Forensic scores. Altman Z''** = 6.56 WC/TA + 3.26 RE/TA + 6.72 EBIT/TA + 1.05 BVE/TL, with distress below 1.1. Z'' was chosen because it needs no market value (unavailable or noisy for distressed and suspended Indian stocks) and no sales/TA (sector-biased), and it is Altman's emerging-market and non-manufacturer form. On the Polish holdout, Z'' alone gives PR-AUC 0.266. **Piotroski F** uses 9 signals; **Beneish M** uses 8 variables and flags above -1.78, with missing indices set to neutral and counted.
- **Composite** = model log-odds + a pre-specified log-odds overlay, which was not fitted: {'piotroski_F<=2': 0.5, 'piotroski_F=3': 0.25, 'beneish_M>-1.78': 0.3, "altman_Z''<1.1": 0.3, 'price_12m<-50%': 0.7, 'price_12m<-30%': 0.35, 'accounts_overdue_18m': 0.5}.
- **Quarterly mapping.** Ratios change once a year, when the annual report becomes public. Price and overdue-accounts signals change quarterly. "4Q" means the score at the last quarter end at least 12 months before the event; "8Q" means at least 24 months.
- **Explanations.** The top 3 positive contributors come from SHAP TreeExplainer log-odds plus the overlay terms, written as plain sentences with each value and its peer percentile.

4. Polish holdout results



threshold	alerts	alert_rate	FN	FP	TP	cost	rule
0.048	1891	0.320	92	1573	318.000	3413.000	bayes_1_over_21
0.040	1893	0.320	92	1575	318.000	3415.000	val_cost_optimal
0.213	341	0.058	270	201	140.000	5601.000	top_5pct_capacity
	0		410	0		8200.000	flag_nobody

5. Indian backtest results

Alert-cap check: the most alerts in any evaluable quarter is 40, and the highest alert rate is 4.99% (45 quarters, rule min(40, floor(0.05 x N))).

5.1 Recall at the capped 5% alert rate (exact 95% CI)

variant	recall@4Q all [95% CI]	recall@4Q evaluable	recall@8Q all	recall@8Q evaluable	flag in [-8Q,-4Q]	peer flag rate
Polish-trained model only	0/15 [0.00-0.22]	0/14 [0.00-0.23]	0/15 [0.00-0.22]	0/12 [0.00-0.27]	0/15	1.1%
+ forensic overlay (Z'', F, M)	1/15 [0.00-0.32]	1/14 [0.00-0.34]	0/15 [0.00-0.22]	0/12 [0.00-0.27]	1/15	1.5%
Full composite (headline)	1/15 [0.00-0.32]	1/14 [0.00-0.34]	0/15 [0.00-0.22]	0/12 [0.00-0.27]	1/15	1.1%
Post-hoc + ICR/Debt-EBITDA rules (in-sample)	1/15 [0.00-0.32]	1/14 [0.00-0.34]	0/15 [0.00-0.22]	0/12 [0.00-0.27]	1/15	0.9%

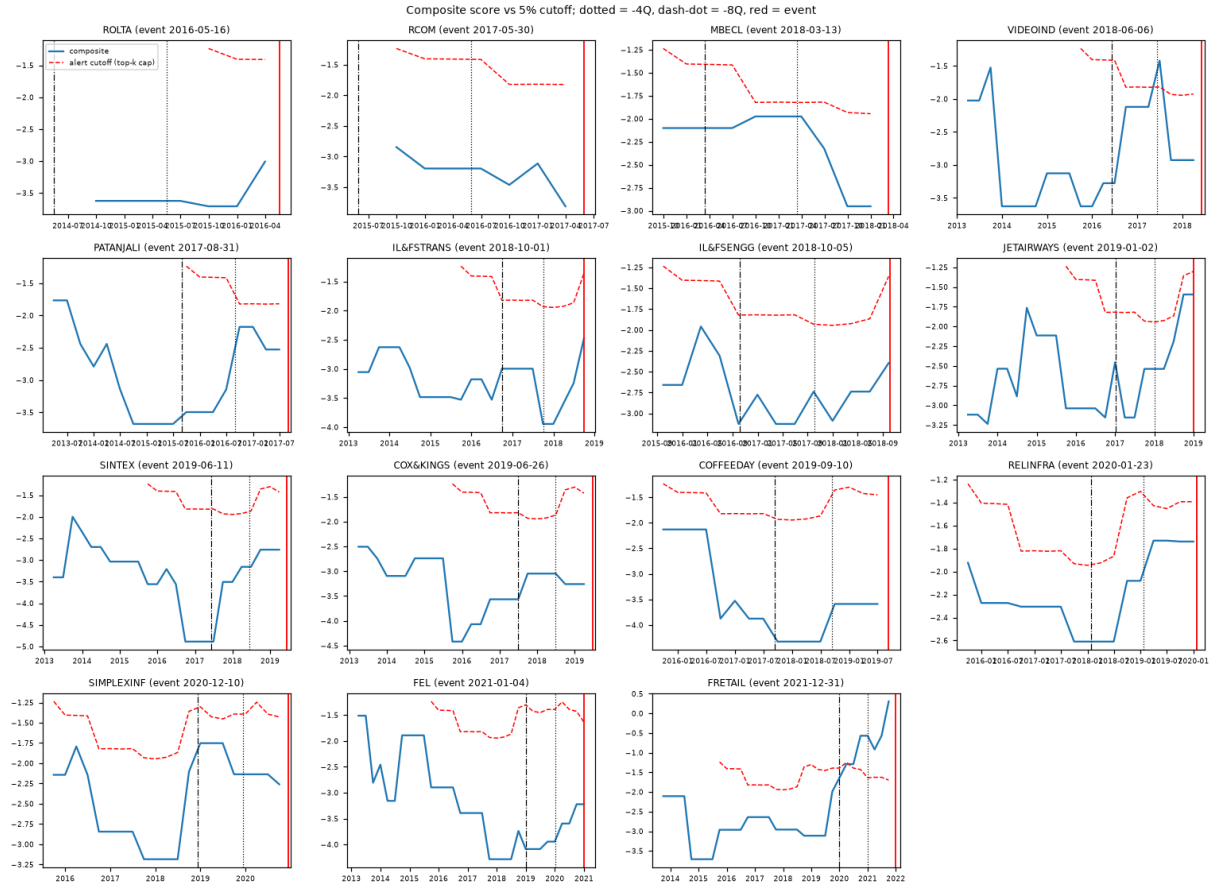
Sensitivities:

run	recall@4Q (full)	recall@8Q (full)	model-only @4Q
headline (lag 183d, full universe)	1/15 [0.00-0.32]	0/15 [0.00-0.22]	0/15 [0.00-0.22]
60-day availability lag	1/15 [0.00-0.32]	0/15 [0.00-0.22]	0/15 [0.00-0.22]
large/mid-cap book only (Nifty Total Market)	1/15 [0.00-0.32]	0/15 [0.00-0.22]	2/15 [0.02-0.41]

5.2 Rank of every case (composite score; rank / N eligible; n/e = before universe coverage)

slug	event_date	q_4Q	fy_used_4Q	rank@4Q	flag 4Q	rank@8Q	flag 8Q	first flag (Q before event)	rank@4Q large-cap book
ROLTA	2016-05-16	2015-03-31	Mar 2014	n/e	False	no data			n/e
RCOM	2017-05-30	2016-03-31	Mar 2015	360/685 (cap 34)	False	no data			222/436
MBECL	2018-03-13	2016-12-31	Mar 2016	52/716 (cap 35)	False	122/685	False		27/457
VIDEOIND	2018-06-06	2017-03-31	Dec 2015	69/716 (cap 35)	False	379/685	False	4.000	38/457
PATANJALI	2017-08-31	2016-06-30	Mar 2015	343/684 (cap 34)	False	n/e	False		210/435
IL&FSTRANS	2018-10-01	2017-09-30	Mar 2017	409/739 (cap 36)	False	186/716	False		253/475
IL&FSENGG	2018-10-05	2017-09-30	Mar 2017	136/739 (cap 36)	False	207/716	False		72/475

slug	event_date	q_4Q	fy_used_4Q	rank@4Q	flag 4Q	rank@8Q	flag 8Q	first flag (Q before event)	rank@4Q large-cap book
JETAIRWAYS	2019-01-02	2017-12-31	Mar 2017	103/739 (cap 36)	False	114/716	False		51/475
SINTEX	2019-06-11	2018-03-31	Mar 2017	223/738 (cap 36)	False	616/716	False		126/474
COX&KINGS	2019-06-26	2018-03-31	Mar 2017	200/738 (cap 36)	False	317/716	False		109/474
COFFEEDAY	2019-09-10	2018-06-30	Mar 2017	512/737 (cap 36)	False	400/715	False		326/473
RELINFRA	2020-01-23	2018-12-31	Mar 2018	96/764 (cap 38)	False	116/739	False		42/496
SIMPLEXINF	2020-12-10	2019-09-30	Mar 2019	103/794 (cap 39)	False	96/766	False		49/522
FEL	2021-01-04	2019-12-31	Mar 2019	462/794 (cap 39)	False	482/764	False		278/522
FRETAIL	2021-12-31	2020-12-31	Mar 2020	2/834 (cap 40)	True	70/794	False	6.000	2/558



Driver examples at 4Q: **ROLTA**: Sales growth (sales / prior-year sales): not reported in the filings (imputed with the training median) - verify manually | (Pre-tax profit + depreciation) / sales: not reported in the filings (imputed with the training median) - verify manually | Long-term liabilities / equity is high at 1.67 (worse than 89% of monitored borrowers); this raises the distress score. **RCOM**: Asset turnover (sales / total assets) is low at 0.27 (worse than 86% of monitored borrowers); this raises the distress score | Sales growth (sales / prior-year sales): not reported in the filings (imputed with the training median) - verify manually | (Net profit + depreciation) / total liabilities is low at 0.08 (worse than 61% of monitored borrowers); this raises the distress score. **MBECL**: Piotroski F-score only 0/9: profitability, cash flow, leverage and efficiency are mostly deteriorating | Asset turnover (sales / total assets) is low at 0.62 (worse than 65% of monitored borrowers); this raises the distress score | (Net profit + depreciation) / total liabilities is low at -0.07 (worse than 96% of monitored borrowers); this raises the distress score. **VIDEOIND**: Asset turnover (sales / total assets) is low at 0.22 (worse than 89% of monitored borrowers); this raises the distress score | Sales growth (sales / prior-year sales): not reported in the filings (imputed with the training median) - verify manually | (Net profit + depreciation) / total liabilities is low at -0.02 (worse than 93% of monitored borrowers); this raises the distress score.

5.3 Researched soft signals (case evidence, not in metrics)

slug	event_date	n_signals	n_before_event	n_at_least_4Q_before	earliest_signal	lead_months_earliest
ROLTA	2016-05-16	1	0	0		
RCOM	2017-05-30	0	0	0		
MBECL	2018-03-13	0	0	0		
VIDEOIND	2018-06-06	1	1	1	2016-06-08	23.900
PATANJALI	2017-08-31	0	0	0		
IL&FSTRANS	2018-10-01	2	2	0	2018-09-10	0.700
IL&FSENGG	2018-10-05	1	1	0	2018-09-04	1.000
JETAIRWAYS	2019-01-02	3	3	0	2018-05-24	7.300
SINTEX	2019-06-11	0	0	0		
COX&KINGS	2019-06-26	1	1	0	2019-06-11	0.500
COFFEEDAY	2019-09-10	1	1	0	2019-06-30	2.400
RELINFRA	2020-01-23	2	2	0	2019-06-14	7.300
SIMPLEXINF	2020-12-10	0	0	0		

slug	event_date	n_signals	n_before_event	n_at_least_4Q_before	earliest_signal	lead_months_earliest
FEL	2021-01-04	0	0	0		
FRETAIL	2021-12-31	3	3	3	2020-04-01	21.000
RELCAPITAL	2019-09-20	2	2	0	2019-04-26	4.800

Only signals dated before the event count as warnings. For example, Rolta's S&P D on 2016-06-20 is after its event and only confirms it.

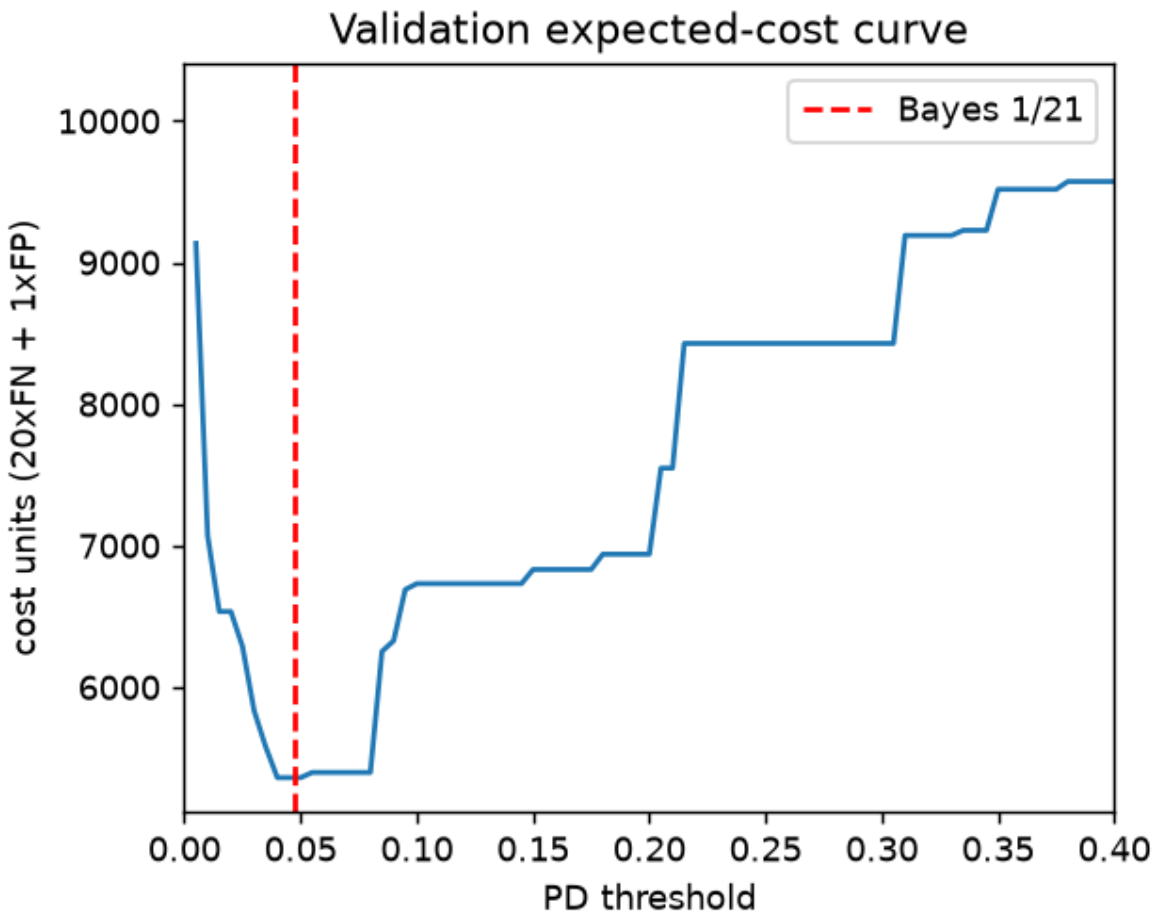
5.4 Cost

Matched case/peer panel, realised cost (missed defaults x 20 + false alarms x 1):

lead	variant	TP	FN	FP	calc
4Q	score_model	0	15	0	20.0 x 15 missed + 1.0 x 0 false alarms = 300.0
4Q	score_model_forensic	1	14	0	20.0 x 14 missed + 1.0 x 0 false alarms = 280.0
4Q	score_full	1	14	0	20.0 x 14 missed + 1.0 x 0 false alarms = 280.0
4Q	score_posthoc_credit	1	14	0	20.0 x 14 missed + 1.0 x 0 false alarms = 280.0
8Q	score_model	0	15	2	20.0 x 15 missed + 1.0 x 2 false alarms = 302.0
8Q	score_model_forensic	0	15	2	20.0 x 15 missed + 1.0 x 2 false alarms = 302.0
8Q	score_full	0	15	2	20.0 x 15 missed + 1.0 x 2 false alarms = 302.0
8Q	score_posthoc_credit	0	15	1	20.0 x 15 missed + 1.0 x 1 false alarms = 301.0

Universe model-expected cost (mean per evaluable quarter). PDs are calibrated on Polish data and prior-shifted to an assumed base rate of 1%, 2.5% or 5% (an assumption, not a sourced figure):

base_rate	policy	alerts	exp_missed_defaults	exp_false_alarms	exp_cost
0.01	bayes_PD>=1/21	45.16	6.85	41.10	178.16
0.01	capped_top_k	38.38	7.32	34.80	181.27
0.01	flag_nobody	0.00	10.90	0.00	218.07
0.03	bayes_PD>=1/21	135.18	10.46	119.97	329.11
0.03	capped_top_k	38.38	18.01	30.72	390.82
0.03	flag_nobody	0.00	25.67	0.00	513.31
0.05	bayes_PD>=1/21	246.87	12.77	211.85	467.28
0.05	capped_top_k	38.38	35.13	25.71	728.28
0.05	flag_nobody	0.00	47.79	0.00	955.86



Operating policy

Book. 930 real listed Indian non-financial companies scored at 2026-09-30 (frozen universe, `data/reference/universe_frozen.csv`; a proxy for the brief's 800-borrower book, not simulated). **Alert threshold.** Each quarter, flag the top k = min(40, floor(5% x N)) by composite score (this quarter: k = 40, 40 alerts). This is a hard cap, so the review team never receives more than 40 names. **Quarterly cycle.** 1. T+5 days: refresh filings and prices. Use only statements public by T (FY end + 183 days). Score every borrower. 2. Flagged ranks 1-10 -> **ESCALATE**: senior credit officer within 2 weeks; covenant test, bank-statement/GST cross-check, management call. 3. Other flagged names -> **REVIEW**: analyst deep-dive this quarter using the three plain-language drivers; confirm or dismiss within 30 days. 4. Ranks up to 80 -> **WATCH**: monthly monitoring of price, ratings and pledges. 5. **Override, escalate regardless of**

rank: downgrade to sub-investment grade or by 2 or more notches; auditor resignation, qualification or disclaimer; results deferred past the LODR deadline; promoter pledge above 50% or up more than 10 pp in a quarter; any lender reporting SMA-1/SMA-2 (CRILC). 6. **Escalation path**: analyst -> credit committee (watch-list tag, limit freeze, extra security) -> CRO/Board risk committee above delegated limits -> RBI June 2019 prudential framework (inter-creditor agreement) on SMA-2 or confirmed stress.

Threshold and cost. A missed default costs 20.0 and a false alarm costs 1.0. Flagging is worth it when (1-p) x 1.0 < p x 20.0, i.e. **p > 1/21 = 0.0476** (Polish validation cost optimum: 0.040). Capacity binds, so the operating rule is "top-k capped at 40; within that, everyone with PD > 1/21 is reviewed first". At an assumed 2.5% annual base rate, model-expected cost this quarter is **353.7** units with capped top-k, versus 451.3 with no reviews and 323.1 for an uncapped Bayes rule (118 alerts, which exceeds capacity).

Latest capped watch-list, top 10. These are screening outputs, not predictions of default:

rank	company	score_full	Z2	F_score	driver_1
1	GSS Infotech Ltd	0.430	4.074	3.000	Receivable days is unusual at 248.86 (98th percentile of monitored borrowers); this raises the distress score
2	Hexa Tradex Ltd	0.127	8.829	4.000	Quick ratio ((current assets - inventory) / short-term liabilities) is low at 0.00 (worse than 100% of monitored borrowers); this raises the distress score
3	Asian Hotels (East) Ltd	-0.362	-1.622	5.000	Asset turnover (sales / total assets) is low at 0.17 (worse than 92% of monitored borrowers); this raises the distress score
4	TECIL Chemical & Hydro Power Ltd	-0.432	-16.843		Asset turnover (sales / total assets) is low at 0.00 (worse than 100% of monitored borrowers); this raises the distress score
5	Mangalam Drugs and Organics Ltd	-0.500	-0.085	3.000	Share price fell 56% over 12 months: the market is pricing serious stress
6	Sobha Ltd	-0.641	2.575	7.000	Asset turnover (sales / total assets) is low at 0.27 (worse than 88% of monitored borrowers); this raises the distress score
7	Blue Coast Hotels Ltd	-0.854	-36.963	4.000	Quick ratio ((current assets - inventory) / short-term liabilities) is low at 0.12 (worse than 99% of monitored borrowers); this raises the distress score
8	Vedanta Iron & Steel Ltd	-0.880	-6.477		Asset turnover (sales / total assets) is low at 0.70 (worse than 55% of monitored borrowers); this raises the distress score
9	TRF Ltd	-0.975	-4.317	2.000	Asset turnover (sales / total assets) is low at 0.24 (worse than 89% of monitored borrowers); this raises the distress score
10	Vardhman Polytex Ltd	-0.980	-14.170	8.000	Quick ratio ((current assets - inventory) / short-term liabilities) is low at 0.16 (worse than 98% of monitored borrowers); this raises the distress score

6. Missing data

slug	n_fy_on_screener	first_fy	n_fy_public_before_event	has_3y_pre_event	missing_feature_share_pre_event	F_score_missing_years	M_score_m
ROLTA	13	Mar 20149m	3	True	0.512	2	3
RCOM	12	Mar 2015	2	False	0.018	1	1
MBECL	12	Mar 2015	3	True	0.012	1	1
VIDEOIND	16	Sep 2006	14	True	0.436	11	11
PATANJALI	12	Mar 2010	7	True	0.005	2	1
IL&FSTRANS	12	Mar 2007	12	True	0.003	2	1
IL&FSENGG	12	Mar 2015	4	True	0.009	1	1
JETAIRWAYS	11	Mar 2008	11	True	0.003	2	1
SINTEX	12	Mar 2011	8	True	0.004	1	1
COX&KINGS	12	Mar 2009	10	True	0.004	2	1
COFFEEDAY	12	Mar 2015	4	True	0.009	1	1
RELINFRA	12	Mar 2015	5	True	0.007	1	1
SIMPLEXINF	12	Mar 2015	6	True	0.006	1	1
FEL	13	Jun 2010	11	True	0.143	4	4
FRETAIL	10	Dec 2012	10	True	0.254	4	4
RELCAPITAL	12	Mar 2013	6	True	0.006	1	1

Videocon used irregular fiscal periods, so growth ratios are blanked for non-12-month periods. Ruchi Soya has a gap between FY2019 and FY2025 on screener.in, and only pre-event years are used. Rolta's statements start in FY2014, so its 8Q point cannot be scored.

7. Survivorship bias, sector effects, limitations

- **Survivorship bias (material).** The universe was built from current listings, the Nifty Total Market and NSE lists as of October 2026. Firms that delisted, were suspended or defaulted between 2015 and 2026 are missing unless they are one of the cases. As a result, (i) historical quarters contain only survivors, so peer and universe "false alarms" are overstated as specificity (some flagged names, such as GTL, were in fact distressed), and (ii) the 5% budget is filled from survivors. A point-in-time universe from historical index constituents or exchange lists is the main fix.
- **Selection.** The cases are famous large defaults, picked because they have documented dates. Small-company defaults, which the model ranks well, are under-represented. With n = 15, the confidence intervals are wide.
- **Sector effects.** Infrastructure, EPC, telecom, airlines and realty structurally carry high leverage and low Z''. The cutoff is universe-wide, so a sector-relative rank is the next step.

- **Transfer gap.** Polish SMEs from 2000 to 2013 are not Indian conglomerates whose subsidiaries rely on group guarantees (IL&FS, ADAG, Future).
- **Annual granularity.** At the 4Q point, the latest statements are often 15-21 months old.
- **Source caveats.** Screener may show restated figures. Event dates are the earliest found, and earlier defaults are possible (Videocon, McNally). The post-hoc credit-rule variant is in-sample.

8. Reproducibility

Run `make setup && make all` (`python run_all.py`), then `make test` (pytest). Start the local app with `make app` (Streamlit). `python run_all.py --scrape` re-fetches source pages; this changes the frozen inputs. Seed 42. No push and no deployment.